Does the Habermas Machine under-represent minority opinions?

A replication of Figure 4C of Tessler et al. (2024)

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September 8, 2026

Introduction

In the original Habermas Machine (HM) paper (Tessler et al., 2024), the authors consider the possibility that the HM might exhibit a "tyranny of the majority" phenomenon, in which minority perspectives are under-represented in the HM's output, relative to their prevalence among the population. They found that it did not do so. To do this, they developed a novel method for describing to what degree a summary statement reflected majority and minority opinions. This finding is significant because the HM's design goal was to produce a consensus statement that all participants would believe represented their opinion. In this report, I aim to replicate this finding using the publicly-released data from the original HM paper. To cut to the chase, I was unable to replicate this finding.

This report assumes familiarity with the Habermas Machine paper, and also largely assumes familiarity with the relevant sections of the Supplementary Materials. I aim to provide a technical report of my findings, rather than writing for a general academic audience.

Approach to reproduction

The authors released granular data from the experiments that they ran. They did not release the code that they used to perform the "tyranny of the majority" claim. I used details in their Supplementary Materials (SM) to attempt to replicate their analysis pipeline. To understand the deviations between the original analysis and my reproduction, I calculated several estimands reported in the original paper. There are several points where the SM are ambiguous and leave an aspect of the analysis pipeline unspecified. I approach this in the following manner: First, I present the analysis pipeline which I believe is most likely to match the original analysis. Then, after I have presented this "most-likely" pipeline, I perform an exhaustive sensitivity analysis, in which I vary these ambiguous details (Sensitivity analysis).

This report's scope is limited to the minority-weight result shown as Figure 4C in the main paper.

What is the Habermas Machine (HM)?

Briefly, the HM is a tool for abstractive summarisation of people's opinions. A group of people each provide their opinion on a controversial topic. The HM iteratively produces and refines a consensus statement summarising the opinions in the group. First, it generates n initial statements. It then ranks these statements according to a simulated vote, in which it attempts to simulate the preferences of each opinion's author across the initial statements. The first-vote winner thus produced is shown to each (human) author. Each author provides feedback. This feedback is provided to the LLM. It generates a second round of candidate statements. These again go through a simulated vote, resulting in the final statement.

The finding that is the focus of this report is that when one compares the extent to which minority opinions are represented in the population (proportion of 0.29), this is roughly the same as how represented minority opinions are in the initial generated statements (0.28), and is <u>less</u> than how well represented minority opinions are in the final statement (0.36).[1] This is shown in Figure 1, which is copied from Figure 4(c) in the paper.

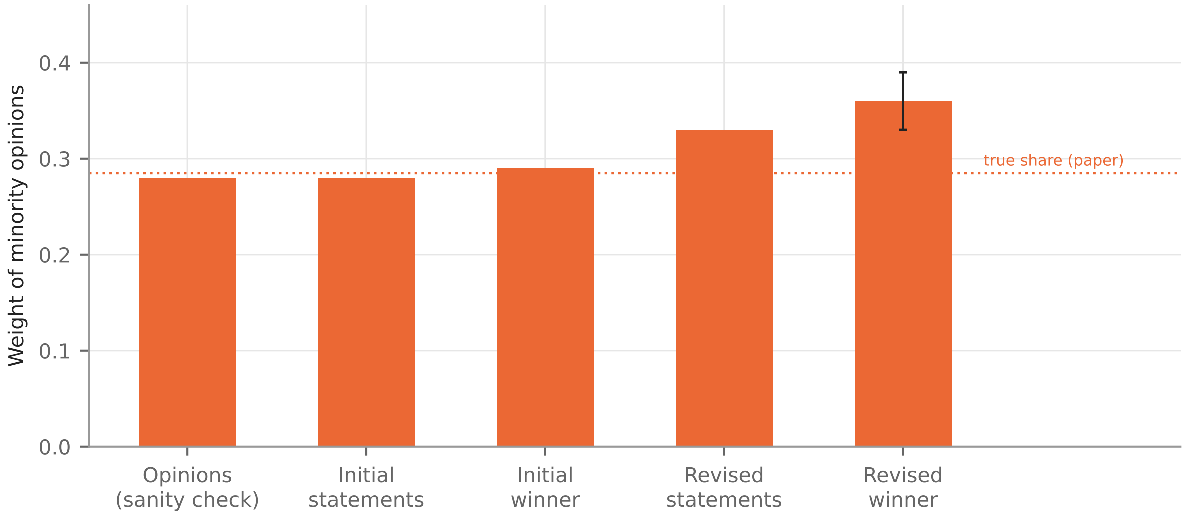

Figure 1: The paper's own result, redrawn from the data labels of SM Figure S60 ("Position Embedding" panel), which SM 5.2.2 identifies as the expanded version of main-text Figure 4C. Dotted line: the true proportion of minority opinions (0.285). Error bar on the revised winner: ±0.03, one standard error of the estimated regression coefficient, as printed in the paper. $n = 1047$ rounds.

What is a minority (or majority) group?

In the papers' experiments, groups of between 4–5 people took part in the deliberative process mediated by the HM. Before writing their free-text opinion, they indicated their level of agreement with a political proposition using a Likert scale of 1–7. It was thus possible to provide a neutral (4) opinion. A group was defined as having a minority group if there was an unequal amount of agreeing (5–7) and disagreeing (1–3) opinions. Groups without a minority group were excluded from this analysis.

The sample for this analysis is the pre-registered groups of cohorts 1–3: 1047 rounds, of which 560 contain a minority under this rule, giving a true minority share of 0.262.[2]

[1] The dotted "true proportion" line in SM Figure S60 is at 0.285; the 0.28 and 0.29 bars are the initial candidates and the initial winner respectively. The comparison in this sentence is therefore between 0.285 and 0.36.

[2] The paper reports 0.285. The difference between our result and the paper's is potentially explained by different handling of neutral (Likert value 4) ratings. SM 5.4.1 counts a neutral rating as part of the non-minority group: "we may define the 'minority' side of the Likert scale as the side that contains the smaller number of opinions, and define all other opinions as 'non-minority' opinions—which include 'majority' opinions [. . .] as well as 'neutral' opinions (i.e. with a Likert rating of 4), if any." Applied literally, with rounds that have equal numbers on each side excluded, that rule gives 0.262 on these rounds. One alternative way to handle neutral opinions is to drop them from their groups. This gives 0.289. We use the first option, because it better matches

How does the HM paper measure how well-represented minority opinions are?

This part of the analysis has two components. First, each participant's free-text opinion and the statements produced by the HM (including intermediate statements produced in its iterative process) are given a scalar score to measure to what extent they agree or disagree with the political proposition. The paper refers to this score as the "position component score".[3]

The position component scores of original human opinions and LLM-generated statements are then used to measure how similar the average position component scores among a minority group are, to the position component score of the LLM-generated statement.

I now detail my reproduction of each of these steps.

Measuring the position component score

To measure the position component score of a human-authored opinion or a LLM-generated statement, it is first mapped into an embedding space, and then projected onto the vector going between two points.[4]

The main detail missing from the SM is what exact embedding model was used for this process. While the SM describes Sentence-T5, there are various sizes of this model that could be used.

Validating the position component score

The original authors validate their position component score method by using the original human responses, comparing the human's self-reported Likert scores against the estimated position component scores derived from the method above. In the original paper, they report an R^2 of 0.41.[5] In our reproduction, the R^2 of each embedding model is shown in Table 6 (Appendix A).

We choose model sentence-t5-large as the most-likely model used in the original analysis, because its marginal r^2 of 0.392 is within 0.018 of the paper's 0.41.

Measuring representativeness

Given a set of human-authored opinions $\{x_i \mid i \in 1, \ldots, n\}$, a partition of opinions into minority M & non-minority N sets, position component scores $\{a_i \mid i \in 1, \ldots, n\}$, the "representa-

the description in the SM, but consider the other option in the sensitivity analysis below.

[3]It is best understood as a way of measuring the level of agreement with the proposition, similar to a political valence score: a single number that places a text between the negating and the affirming position on the question.

[4]SM 5.1.2 gives the method for constructing the two endpoints — a generic prefix ("Yes, I agree." or "No, I disagree.") concatenated with the question's own affirming or negating position statement — and one illustrative example, but not the exact phrases used for each question. Here each endpoint is the prefix followed by the position statement released with the data; the axis is the unit vector from the negating to the affirming endpoint (SM eq. 3) and the score is the projection onto it (SM eq. 5). The readings that description admits are tested in the sensitivity analysis (Table 5).

[5]The paper does not say exactly how this R^2 is computed. Appendix A sets out the candidate definitions, how I settled on one, the regression that produces it, and its value under each embedding model.

tion" of an opinion x_i in a target output x_0 (with position component score a_0) is given by λ_i, where λ_i is part of the solution to the optimisation problem

$$\text{set } \lambda_i \text{ to minimise } \left(a_0 - \sum_i \lambda_i a_i\right)^2 \qquad \text{such that} \qquad \sum_i \lambda_i = 1 \;\; \& \;\; \lambda_i \geq 0.$$

To calculate the representation of M & N, simply calculate the average representation of their elements: $\lambda_M := \frac{1}{|M|} \sum_{x_i \in M} \lambda_i$.

The implementation check

To test that the implementation of this procedure works as expected, the original authors ran a check where each human opinion in turn is treated as the target and regressed on convex combinations of that same set of opinions. Averaged over the set, the recovered minority weight should come back as simply the proportion of opinions in the minority. We replicate this implementation check. It passes: the recovered weight is 0.263 against a true share of 0.262; the paper reports 0.28 against its true share of 0.285. This is the leftmost bar of Figure 2.

Results

We fail to replicate the results from the original paper. Figure 2 displays an adjusted version of Figure 1 above, where we contrast the original paper's results against our own.[6]

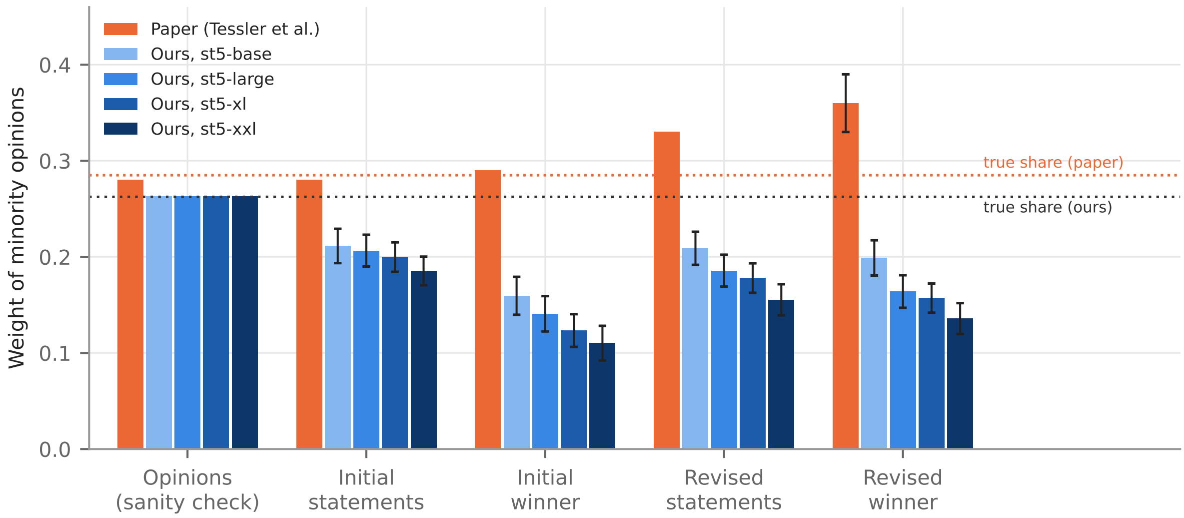

Figure 2: Minority weight at each phase: the paper against this reproduction under each Sentence-T5 size. Error bars: paper, ±1 standard error of the regression coefficient as printed (SM Figure S60); ours, ±1 cluster-bootstrap standard error over rounds. The two dotted lines are the true minority share in each case (0.285 for the paper, 0.262 here; the difference is the treatment of neutral opinions). $n = 560$ rounds with a minority, from 1047 pre-registered rounds in cohorts 1–3, for every model.

Three observations:

[6]The paper's error bar is ±1 standard error of the estimated regression coefficient, as SM Figure S60 states. Error bars on our estimates are ±1 standard error from a cluster bootstrap over rounds (500 resamples, drawing the same rounds in every phase, so that differences between phases are bootstrapped on matched samples). We take a bootstrap approach because it is unclear how the original authors calculate standard errors in the convex regression setting.

1. The sanity check reproduces (as described earlier).
2. Every statement phase sits below the true share rather than at or above it: under sentence-t5-large the revised winner carries 0.164 ± 0.017 against the paper's 0.36.
3. In the paper the four phases rise monotonically, whereas here the winner-selection step *drops* minority weight at both rounds (initial candidates $0.206 \rightarrow$ initial winner 0.141), and the revision step partially restores it (0.164). The paper's rise from initial winner to revised winner is therefore present (+0.023, 95% CI $[-0.001, 0.044]$, $p = 0.030$), but it starts from a lower base and does not carry the final statement above proportionality; measured from the initial candidates instead, the change over the whole process is -0.043 (95% CI $[-0.077, -0.008]$).

The same holds at every model size (Table 1): the minority weight of the revised winner falls as the model grows (0.199, 0.164, 0.157 and 0.136 for base, large, xl and xxl) while the quality of the position axis rises (Table 6).

Table 1: Minority weight at each phase by embedding model, with the cluster-bootstrap standard error in parentheses; the paper's values are from SM Figure S60. The sanity-check row regresses the opinions on themselves and should recover the true share (0.285 in the paper, 0.262 here).

	paper	base	large	xl	xxl
Opinions (sanity check)	0.28	0.263	0.263	0.263	0.263
Initial statements	0.28	0.211 (0.018)	0.206 (0.017)	0.200 (0.015)	0.185 (0.015)
Initial winner	0.29	0.160 (0.020)	0.141 (0.018)	0.123 (0.017)	0.110 (0.018)
Revised statements	0.33	0.209 (0.017)	0.186 (0.017)	0.178 (0.015)	0.155 (0.016)
Revised winner	0.36 (0.03)	0.199 (0.018)	0.164 (0.017)	0.157 (0.015)	0.136 (0.016)

Validation checks

Table 2 has several validation checks where we compared intermediate parameter estimates against those reported in the paper.

Table 2: Checks on the reproduction, independent of the minority-weight result: the sample reconstructed from the released data against the paper's description of it, and the intermediate estimates the paper reports.

check	paper	ours (sentence-t5-large)
Pre-registered groups in cohorts 1–3 (SM 4.4)	349	349
Unique participants in those groups (SM 4.4)	1692	1720
Rounds (3 questions per group)	1047	1047
Rounds with a minority, entering the regression	not stated	560
Opinions in the position-axis regression (Appendix A)	not stated	4979
Sanity check: opinions regressed on themselves recover the minority share	0.28 vs 0.285	0.263 vs 0.262
Position axis quality: marginal r^2 of Likert on position component score	0.41	0.392
Fig. 4A: correlation of position component score with Likert rating	0.64	0.63
Fig. 4B: statements falling within the range of their group's opinions	0.96	0.86

Sensitivity analysis

As noted above, there are several aspects of the analysis plan which are not specified in the SM. These are:

Table 3: Underspecified choices, the options considered, and the option taken in the primary specification above.

ambiguity	possible options	option chosen above
ST5 model size	base, large, xl, xxl	I focus on values from the large model (Appendix A), but Figures 2 and 3 show results across all model sizes
Neutral (Likert 4) opinions	count as non-minority; drop the opinion	count as non-minority (SM 5.4.1)
Basis for splitting minority / non-minority	Likert rating; sign of the position component score (SM Fig. S62)	Likert rating (SM Fig. S60)
Position-axis endpoints: the released statements write the negation as "NOT"	as released; "NOT" lowercased, as in the SM example	as released (Table 5)
Column order: how a round's opinions are assigned to the regressors	order in the released data; random	data order

To explain the last row 'Column order': The regression at each level of division fits one coefficient per column of the design matrix, and each row of that matrix is one round's opinion scores, so a column pools "the j-th opinion" across rounds. It is not described in the SM how datapoints are assigned to columns. It seems self-evident that minority and majority opinions should always stay in their own sets of columns (to prevent them being mixed together in the regression). However, what determines the ordering within each group of columns is unclear; moreover, the results shift slightly when one perturbs the ordering. In Appendix B, I consider a simpler and more robust method which is ordering-invariant (fitting one parameter for all minority opinions, rather than for each column), and find almost identical results.

Endpoint construction

SM 5.1.2 pins the endpoints to "a generic + question-specific combination" and gives the example "Yes, I agree. It is the government's role to [...]"; SM 5.1.1 fixes the axis as the unit vector from the negating to the affirming embedding (eq. 3) and the score as the projection onto it (eq. 5). Each endpoint is therefore a single text: the generic phrase ("Yes, I agree." or "No, I disagree.") followed by the question's affirming or negating position statement as released with the data. Table 4 works through the construction for two questions, the two asked in the most pre-registered rounds of cohorts 1–3, and lists the complete set of endpoint texts that results.

One detail of that description can be read more than one way. The released position statements write the negation in capitals ("...should NOT be ... "), where the SM's example reads "is not". Table 5 runs the primary specification under sentence-t5-large with the statements as released (the pinned reading) and with "NOT" lowercased. The two readings agree to within one bootstrap standard error of one another at every phase, and neither moves the revised winner (0.164 and 0.167) towards the paper's 0.36: the endpoint construction is not where the gap comes from.

Table 4: Two worked examples of the endpoint construction: the questions asked in the most pre-registered rounds of cohorts 1–3 (7 and 7 rounds), their affirming and negating position statements as released with the data, and the complete set of endpoint texts that each reading in Table 5 embeds. Where a statement contains no "NOT" the two readings coincide.

	example 1	example 2
Question	Should the UK raise taxes on people earning more than $200,000 to help pay for the National Health Service?	Should employers be forced to provide their employees with at least one day off from work each week?
Released affirming statement	The UK should raise taxes on people earning more than $200,000 to help pay for the National Health Service.	Employers should be forced to provide their employees with at least one day off from work each week.
Released negating statement	The UK should NOT raise taxes on people earning more than $200,000 to help pay for the National Health Service.	Employers should NOT be forced to provide their employees with at least one day off from work each week.
Pinned reading: statements as released		
Affirming endpoint	"Yes, I agree. The UK should raise taxes on people earning more than $200,000 to help pay for the National Health Service."	"Yes, I agree. Employers should be forced to provide their employees with at least one day off from work each week."
Negating endpoint	"No, I disagree. The UK should NOT raise taxes on people earning more than $200,000 to help pay for the National Health Service."	"No, I disagree. Employers should NOT be forced to provide their employees with at least one day off from work each week."
"NOT" lowercased, as in the SM example		
Affirming endpoint	"Yes, I agree. The UK should raise taxes on people earning more than $200,000 to help pay for the National Health Service."	"Yes, I agree. Employers should be forced to provide their employees with at least one day off from work each week."
Negating endpoint	"No, I disagree. The UK should not raise taxes on people earning more than $200,000 to help pay for the National Health Service."	"No, I disagree. Employers should not be forced to provide their employees with at least one day off from work each week."

Table 5: The primary specification under sentence-t5-large under each reading of the released "NOT": minority weight by phase (cluster-bootstrap standard error in parentheses) and the two axis-quality checks (Pearson r of Figure 4A, marginal r^2 of Appendix A). $n = 560$ rounds with a minority in both columns.

	statements as released (pinned)	"NOT" lowercased, as in the SM example
Initial statements	0.206 (0.017)	0.208 (0.017)
Initial winner	0.141 (0.018)	0.143 (0.019)
Revised statements	0.186 (0.017)	0.187 (0.017)
Revised winner	0.164 (0.017)	0.167 (0.017)
Fig. 4A r	0.63	0.63
Marginal r^2	0.392	0.391

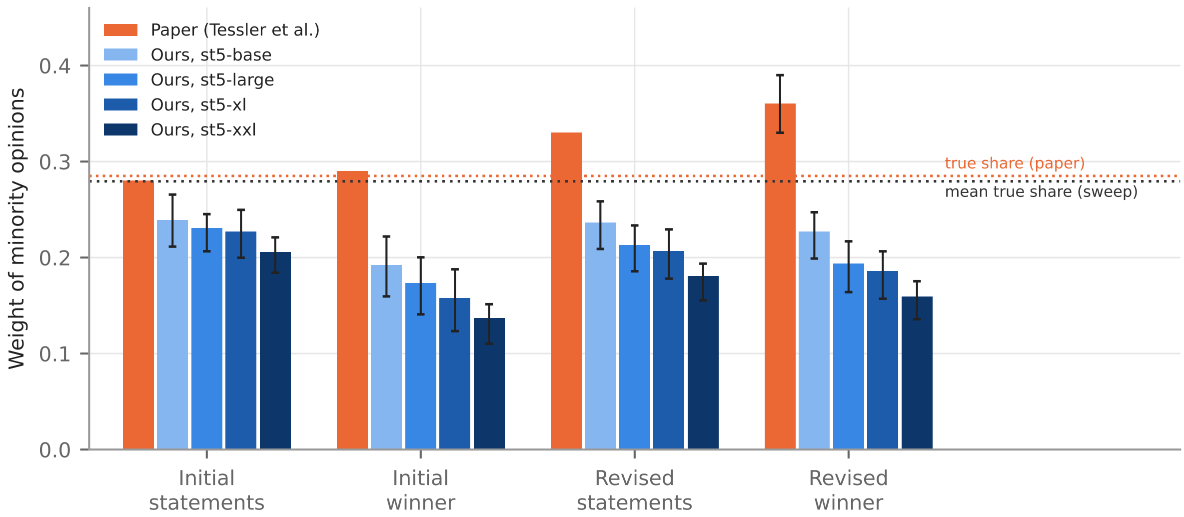

Figure 3: The paper's result against the sensitivity sweep. Orange: the paper's values (SM Figure S60), with its ± 1 standard error on the revised winner. Blue: the mean minority weight at each phase across the 6 specifications run under each embedding model, with whiskers spanning the full range (minimum to maximum) of those specifications. Dotted lines: the paper's true minority share (0.285) and the mean true minority share across all 24 runs (0.279).

The sweep

I reran the analysis with every combination of these decisions (24 runs).[7] Across the runs, I calculated the following metrics:

- Proportion of runs where minority representation increases from initial to final: 0%
- Proportion of runs where minority representation is larger in final than the prevalence among population (0.262):[8] 0%. The largest revised-winner weight in the sweep is 0.247, against a mean true share of 0.279.
- Average representation at each of the 4 stages, with the range across runs: shown in Figure 3.

Discussion

If the above reproduction of the analysis pipeline is correct, then it would appear that the HM actually suppresses minority opinions, if we follow the original paper's interpretation of the above measurement method.

Notably, we are able to reproduce several parameter estimates from the paper, including the validation of the embedding-based position component score and the implementation check of the regression method. We fail to replicate the substantive findings for this part of the paper, however.

[7] 4 model sizes $\times$ 2 neutral-opinion treatments $\times$ 2 column orders when the split is by Likert rating, plus 4 model sizes $\times$ 2 column orders when the split is by the sign of the score, where there are no neutral opinions: 24 runs, 6 per model.

[8] Each run is compared against its own true share rather than a single fixed number, since the minority rule changes the share as well as the weight.

This finding actually aligns somewhat with the finding that people's beliefs shifted towards the majority position. An autonomously Claude-generated replication of Figure 4(d) found a correlation between majority-dominated groups and shifts towards majority opinion (unlike the paper, which found no correlation); however I have not checked this closely yet.

References

Tessler, M. H., Bakker, M. A., Jarrett, D., Sheahan, H., Chadwick, M. J., Koster, R., Evans, G., Campbell-Gillingham, L., Collins, T., Parkes, D. C., Botvinick, M., & Summerfield, C. (2024). AI can help humans find common ground in democratic deliberation. *Science*, *386*(6719), eadq2852. https://doi.org/10.1126/science.adq2852

Appendix A: the paper's R^2 statistic

Note: this section was generated using Claude Fable 5.1 without supervision or code review.

What the paper says. The main text reports: "A random-effects estimator of the regression coefficient for Likert ratings on position embeddings yields a marginal r-squared value of 0.41 (SM 5.1.3)." SM 5.1.3 adds that a model taking the top ten residual embedding components as additional inputs has marginal and conditional r^2 of 0.41 and 0.45, "similar to the model containing only the position component". That leaves three things unspecified: which R^2 is meant — the marginal R^2 of a mixed model, or the plain Pearson r^2 of the scatter in Figure 4A, whose $r = 0.64$ squares to 0.41; what the random effect is grouped by (question, round or participant); and whether only the intercept or also the slope varies by group.

How I settled on a definition. "Marginal" and "conditional" r^2 are the standard names for the Nakagawa–Schielzeth decomposition of variance explained in a mixed model, and the main text names a random-effects estimator, so I take the statistic to be the marginal R^2 of a random-intercept model. I group by round, the unit at which a group forms and a question is answered, and let only the intercept vary, since a random slope would add a further unstated choice. Because the paper's Pearson r^2 and its marginal R^2 coincide at 0.41, the paper's own number cannot separate the two definitions, so Table 6 reports both. Under either definition the ranking of the embedding models, and with it the choice of sentence-t5-large, is the same.

The regression. For participant j in round i, with Likert rating y_{ij} and position component score s_{ij},

$$y_{ij} = \alpha + \beta\, s_{ij} + u_i + \varepsilon_{ij}, \qquad u_i \sim \mathcal{N}(0, \sigma_u^2), \quad \varepsilon_{ij} \sim \mathcal{N}(0, \sigma_\varepsilon^2),$$

fitted by restricted maximum likelihood on the 4979 opinions of the 1047 pre-registered rounds of cohorts 1–3. Writing σ_f^2 for the variance of the fixed-effect predictions $\hat{\beta} s_{ij}$,

$$R^2_{\text{marginal}} = \frac{\sigma_f^2}{\sigma_f^2 + \sigma_u^2 + \sigma_\varepsilon^2}, \qquad R^2_{\text{conditional}} = \frac{\sigma_f^2 + \sigma_u^2}{\sigma_f^2 + \sigma_u^2 + \sigma_\varepsilon^2}.$$

The fitted slope $\hat{\beta}$ under each model is given in Table 6.

Table 6: Fit of the position component score to the Likert rating, by embedding model: marginal and conditional r^2 of the random-intercept model above, the Pearson correlation of the raw scatter (the main-text Figure 4A quantity) with its square, and the fitted slope.

	marginal r^2	conditional r^2	Pearson r	Pearson r^2	$\hat{\beta}$ (SE)
sentence-t5-base	0.291	0.443	0.561	0.315	11.042 (0.259)
sentence-t5-large	0.392	0.505	0.633	0.400	10.686 (0.204)
sentence-t5-xl	0.447	0.530	0.669	0.448	9.967 (0.171)
sentence-t5-xxl	0.503	0.560	0.713	0.509	10.669 (0.161)
paper	0.41	0.45	0.64	0.41	—

Note that a better axis does not bring the minority weight closer to the paper's value; it moves it further away (Results, Table 1).

Appendix B: one weight per block

Note: the text section was generated using Claude Opus 5 and then edited, but I came up with the idea.

The regression in the Results fits one coefficient per column of the design matrix. Each row of that matrix holds one round's opinion scores, so a column pools "the j-th opinion" across rounds, and the j-th opinion is a different person in every round. SM eq. 13 indexes a round's opinions by j without saying how j is assigned. Notably, this means that changing the ordering of data changes results; I verified this empirically. This seems like a flaw in the regression method, but this is the approach that I think best matches the approach described in the paper/SM.

I developed a simpler approach which is ordering-invariant, and tested whether this different approach changed the results or conclusion of the replication.

My idea is the following: instead of calculating column-level regression parameters and then averaging them across minority and non-minority groups, why not just calculate the average position embedding score for minority and non-minority groups, and calculate a single regression parameter for the minority group?

Write $\bar{a}_M$ and $\bar{a}_N$ for the mean position component scores of the round's minority and non-minority opinion groups, and a for the weight given to the minority group. The problem of the Results section then has a single free parameter:

$$\text{set } a \text{ to minimise } \Big(a_0 - a\,\bar{a}_M - (1-a)\,\bar{a}_N\Big)^2 \qquad \text{such that} \qquad 0 \leq a \leq 1.$$

Each of the $|M|$ minority opinions now carries $\lambda_i = a/|M|$ and each of the $|N|$ others $(1-a)/|N|$, so the weight the fit puts on the minority group, $\sum_{x_i \in M} \lambda_i$, is exactly a. This is the same estimand as before with one assumption added — exchangeability within minority/majority opinion groups — and not a different quantity. Nothing else changes: the same levels of division, the same minimum number of rounds per level, the same weighting of levels by rounds and the same cluster bootstrap, so the two fits are computed on identical samples.

A minority / majority mean does not depend on the order of the opinions inside the block, so this fit is invariant to the column order by construction. It is also invariant in fact: across the sanity check and the four phases, under every embedding model, the largest absolute difference between the data order and a random order is 0.

Table 7: Minority weight at each phase with one weight per block, by embedding model, with the cluster-bootstrap standard error in parentheses. The last column repeats the per-opinion fit of Table 1 under sentence-t5-large. The sanity-check row regresses the opinions on themselves and recovers the true share (0.262) under this design as it does under the other; it is not bootstrapped.

	one weight per block				per opinion
	base	large	xl	xxl	large
Opinions (sanity check)	0.263	0.263	0.263	0.263	0.263
Initial statements	0.214 (0.018)	0.208 (0.016)	0.200 (0.016)	0.185 (0.015)	0.206
Initial winner	0.162 (0.020)	0.141 (0.018)	0.125 (0.017)	0.113 (0.018)	0.141
Revised statements	0.211 (0.017)	0.185 (0.016)	0.177 (0.015)	0.157 (0.016)	0.186
Revised winner	0.202 (0.018)	0.163 (0.017)	0.156 (0.015)	0.136 (0.016)	0.164

The parameter estimates are almost exactly identical to the ordering-sensitive approach taken in the paper. This plausibly more robust approach thus produces the same results as the replication.